

HE2003 Econometrics I

Tutorial 2 (Week 3) – Problems & Solutions

Academic Year 2026/2027, Semester 1

Quantitative Research Society @NTU

September 28, 2026

Overview

- **Lecture coverage:** Lecture 2 — *Linear Regression with One Regressor 1*: the simple linear regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$; the OLS minimisation problem; the first-order-condition (FOC) derivation of the closed-form OLS estimators; the summation identities that convert the “ $1/n$ ” form of $\hat{\beta}_1$ into the deviation-from-mean form; and the OLS predicted values and residuals.
- **Questions:** 5 questions from Tutorial Question Set No. 2; solutions extend the official Tut 2 slides, which for Question 2 contain no worked answer at all (see the *Beyond the slides* note there).
- **Notation:** as in lectures — true unknown parameters β_0, β_1 ; generic candidate values b_0, b_1 (the arguments of the minimisation); OLS estimators $\hat{\beta}_0, \hat{\beta}_1$ (random variables, functions of the sample); true error u_i ; OLS residual $\hat{u}_i$ (never e_i); ESS explained, SSR residual, $TSS = ESS + SSR$.

Useful rules (summation notation, restated verbatim from the question sheet).

Let $x_1, \dots, x_n$ and $y_1, \dots, y_n$ be two sequences of numbers and let a be a constant. Then

$$\sum_{i=1}^n ax_i = a \sum_{i=1}^n x_i, \quad (\text{R1})$$

$$\sum_{i=1}^n (x_i + y_i) = \sum_{i=1}^n x_i + \sum_{i=1}^n y_i, \quad (\text{R2})$$

$$\sum_{i=1}^n a = n \cdot a. \quad (\text{R3})$$

These three rules are all that is needed to expand every square and cross-term appearing below.

Question 1 (The idea of OLS estimation and the minimisation problem)

Problem

Explain the idea of OLS estimation and the related minimisation problem. (*Tutorial Question Set No. 2, Q1.*)

Solution

Method 1: From the population model to the least-squares objective

Step 1 — the population model. The course works with the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, 2, \dots, n,$$

where Y_i is the dependent (explained) variable, X_i is the independent (explanatory) variable or regressor, and u_i is the error term. The line $\beta_0 + \beta_1 X$ is the *population regression line*: β_0 is its intercept and β_1 its slope. In practice β_0, β_1 , and hence the population regression line itself, are *unknown* — they are fixed numbers describing the population, not something we can read off directly. We only observe a sample (X_i, Y_i) , $i = 1, \dots, n$, and must use that data to *estimate* β_0 and β_1 .

Step 2 — a candidate line and its prediction error. Fix any two numbers b_0, b_1 (candidate values for the intercept and slope; they need not be the true β_0, β_1). The candidate line $b_0 + b_1 X$ predicts Y_i by $b_0 + b_1 X_i$, so the *prediction error* it makes on observation i is

$$Y_i - (b_0 + b_1 X_i), \quad i = 1, \dots, n.$$

A good line should make these prediction errors small, but they can be positive or negative (points above or below the line), so simply adding them up (which is exactly 0 once b_0, b_1 are chosen optimally, by the first normal equation in Question 3, i.e. $\sum_{i=1}^n \hat{u}_i = 0$) would hide how far the line misses the data. OLS instead measures overall closeness by the **sum of squared errors**,

$$S(b_0, b_1) := \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2 = [Y_1 - (b_0 + b_1 X_1)]^2 + \dots + [Y_n - (b_0 + b_1 X_n)]^2.$$

Squaring makes every term non-negative (so errors cannot cancel) and penalises large misses more than small ones.

Step 3 — the minimisation problem. The **Ordinary Least Squares (OLS) estimators** $\hat{\beta}_0, \hat{\beta}_1$ are defined as the candidate values (b_0, b_1) that make $S(b_0, b_1)$ as small as possible:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{b_0, b_1} \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2.$$

Intuitively, we search over *every* possible straight line (every pair (b_0, b_1)) and pick the one whose sum of squared vertical distances to the data points is smallest. In practice this

search is not done by brute force: we solve the first-order conditions of the minimisation problem (Question 3 carries this out in full), which yields two linear equations in b_0, b_1 with a unique closed-form solution.

Figure. Figure 1 redraws this idea: six data points, one candidate line $b_0 + b_1X$, and the six vertical prediction errors $Y_i - (b_0 + b_1X_i)$ whose squares OLS adds up and minimises.

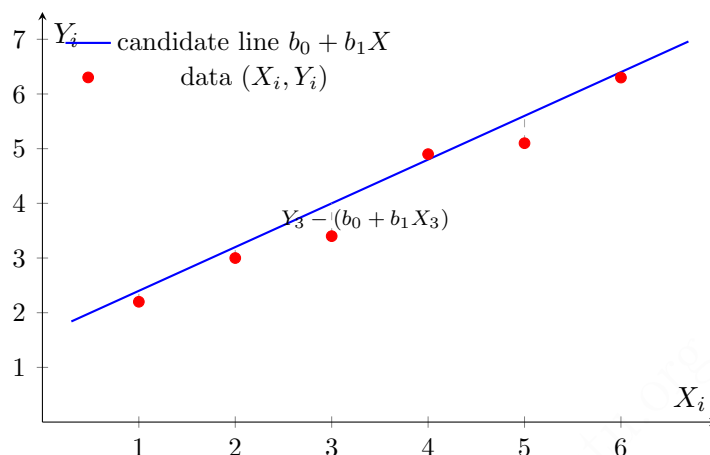


Figure 1. The OLS idea: a candidate line $b_0 + b_1X$ and the vertical prediction errors $Y_i - (b_0 + b_1X_i)$ whose sum of squares $S(b_0, b_1)$ is minimised over (b_0, b_1) . The dashed segments show one such error at each X_i ; note some points lie above the line (positive error) and some below (negative error) — squaring removes the sign before summing.

Interpretation. The OLS line is the “best-fitting” straight line through the scatter of data in exactly the least-squares sense: no other line makes the total squared vertical distance to the data points smaller. This is a purely descriptive/statistical criterion — it says nothing yet about whether $\hat{\beta}_0, \hat{\beta}_1$ are good estimates of the *true* β_0, β_1 ; that question is addressed once the Least Squares Assumptions are imposed (Lecture 3) and unbiasedness/efficiency are established (Lectures 4–5).

Beyond the slides. The official Tut 2 slide states the minimisation problem in words and symbols only, with no accompanying diagram. Figure 1 above supplies the visual that most textbook treatments pair with this definition, making explicit which distances are being squared and summed.

Technique note. Whenever a tutorial or exam question asks you to “explain the idea of OLS,” the three ingredients above are non-negotiable: (i) the population model with *unknown* β_0, β_1 ; (ii) a *generic* candidate line $b_0 + b_1X$ and its prediction error; (iii) the minimisation problem itself, stated with $\arg \min$ over (b_0, b_1) . Question 2 below turns this same distinction (true parameter vs. candidate value vs. estimator) into an explicit three-way comparison.

Question 2 (Notation: (β_0, β_1) vs. (b_0, b_1) vs. $(\hat{\beta}_0, \hat{\beta}_1)$)

Problem

Explain the difference of the notations (β_0, β_1) , (b_0, b_1) , and $(\hat{\beta}_0, \hat{\beta}_1)$ used in class. (*Tutorial Question Set No. 2, Q2.*)

Solution

Beyond the slides. All four slides of the official Tut 2 deck that are devoted to this question contain only the restated question text, repeated verbatim slide after slide, with *no visible answer* — the explanation was evidently given verbally in the tutorial and never transcribed. What follows is a complete answer built from how the three symbols are actually used everywhere else in Lecture 2 (the population model, the minimisation problem, and the closed-form estimator), since that usage pins down exactly what each notation means.

Method 1: Walk through where each symbol first appears

(β_0, β_1) — **the true, unknown population parameters.** These are the intercept and slope of the *population* regression line in the model

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n.$$

β_0, β_1 are two fixed (non-random) numbers that describe the entire population from which the sample is drawn. They are *never observed*: no amount of data collection reveals β_0, β_1 exactly, because the population regression line is a theoretical object, and every individual observation also contains the unobserved error u_i . β_0, β_1 are the targets that estimation tries to approximate; they do not change from sample to sample, because they belong to the population, not to any particular sample.

(b_0, b_1) — **generic candidate values / arguments of the objective function.** These are *placeholder* symbols: any pair of numbers we might propose as a guess for the intercept and slope. They are the variables over which the sum-of-squares function is minimised,

$$S(b_0, b_1) = \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2, \quad \min_{b_0, b_1} S(b_0, b_1).$$

(b_0, b_1) have no statistical meaning by themselves — they are simply the two arguments of the minimisation problem, ranging over all of $\mathbb{R}^2$ while we search for the best-fitting line. Once the search is carried out and the minimum is located, the particular (b_0, b_1) that achieves it is renamed $(\hat{\beta}_0, \hat{\beta}_1)$.

$(\hat{\beta}_0, \hat{\beta}_1)$ — **the OLS estimators: the minimisers, and hence functions of the data.** These are the values of (b_0, b_1) that solve the minimisation problem in Question 1, i.e. that minimise $S(b_0, b_1)$. Because the data $(X_i, Y_i)_{i=1}^n$ that enter $S(\cdot, \cdot)$ are themselves a random sample, the minimisers $\hat{\beta}_0, \hat{\beta}_1$ inherit that randomness: they are *random variables*, computed by a fixed formula (derived in Question 3) applied to the sample. If a different random sample of n classrooms (or firms, or individuals) had been drawn, the

same formula would in general produce different numerical values of $\hat{\beta}_0, \hat{\beta}_1$ — this is the meaning of “ $\hat{\beta}_0, \hat{\beta}_1$ vary from sample to sample,” which underlies their having a sampling distribution (Lectures 4–6). When the formula is evaluated on one *particular*, already-collected sample, the resulting numbers (e.g. $\hat{\beta}_0 = 640.3$, $\hat{\beta}_1 = -4.93$ in Question 5) are called the **estimates** — realised values of the random variables $\hat{\beta}_0, \hat{\beta}_1$.

Method 2: A comparison table

The three layers can be summarised side by side.

Symbol	What it is	Key property
(β_0, β_1)	True population intercept and slope, in $Y_i = \beta_0 + \beta_1 X_i + u_i$.	Fixed, unknown constants; never directly observed; do not vary across samples.
(b_0, b_1)	Generic candidate values; the arguments of the OLS objective $S(b_0, b_1)$.	Free variables ranging over $\mathbb{R}^2$ while minimising; no statistical content on their own.
$(\hat{\beta}_0, \hat{\beta}_1)$	OLS estimators: the particular (b_0, b_1) that minimise $S(b_0, b_1)$.	Functions of the sample data $\Rightarrow$ random variables; vary from sample to sample; a numerical evaluation on one sample is called an <i>estimate</i> .

Interpretation. The three symbols mark three different stages of the same idea: β_0, β_1 is *what we want to know* (a fact about the population); b_0, b_1 is *what we are willing to try* (any candidate line, used purely to set up the optimisation); and $\hat{\beta}_0, \hat{\beta}_1$ is *what OLS gives back* (the best candidate, computed from data, hence itself uncertain). Confusing β_1 with $\hat{\beta}_1$ is a common exam slip: β_1 is a single unknown number, while $\hat{\beta}_1$ is a random variable whose realised value (the “estimate”) is what regression output such as STATA actually prints.

Technique note. A quick way to keep the three straight: β is Greek and unhatted $\rightarrow$ population truth; a plain Roman letter $b \rightarrow$ a dummy/candidate argument inside a min or arg min; a Greek letter with a hat, $\hat{\beta}$, $\rightarrow$ “the number/formula that came out of solving the estimation problem,” which is why *every* estimator in this course (later also $\hat{\sigma}_{\hat{\beta}_1}^2$, $\text{SE}(\hat{\beta}_1)$, $\hat{u}_i$) carries a hat.

Question 3 (Deriving the OLS estimators from the FOCs)

Problem

Derive the formula of the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ following the steps in the lecture note. (*Tutorial Question Set No. 2, Q3.*)

Solution

Method 1: First-order conditions, solved in full

Step 1 — the objective. By Question 1, the OLS estimators solve

$$\min_{b_0, b_1} S(b_0, b_1), \quad S(b_0, b_1) = \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2.$$

Step 2 — both partial derivatives, via the chain rule. Let $g_i = Y_i - b_0 - b_1 X_i$, so that $S(b_0, b_1) = \sum_{i=1}^n g_i^2$. By the chain rule, for each i ,

$$\frac{\partial(g_i^2)}{\partial b_0} = 2g_i \cdot \frac{\partial g_i}{\partial b_0} = 2g_i \cdot (-1), \quad \frac{\partial(g_i^2)}{\partial b_1} = 2g_i \cdot \frac{\partial g_i}{\partial b_1} = 2g_i \cdot (-X_i).$$

Summing over i ,

$$\frac{\partial S}{\partial b_0} = \sum_{i=1}^n 2(Y_i - b_0 - b_1 X_i)(-1) = -2 \sum_{i=1}^n (Y_i - b_0 - b_1 X_i), \quad (\text{P0})$$

$$\frac{\partial S}{\partial b_1} = \sum_{i=1}^n 2(Y_i - b_0 - b_1 X_i)(-X_i) = -2 \sum_{i=1}^n (Y_i - b_0 - b_1 X_i) X_i. \quad (\text{P1})$$

Step 3 — normal equations. The minimisers $(\hat{\beta}_0, \hat{\beta}_1)$ satisfy (P0)=0 and (P1)=0; dividing each by -2 and then by n gives the two *normal equations*

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0 \quad \implies \quad \bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X} = 0, \quad (4)$$

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0 \quad \implies \quad \frac{1}{n} \sum_{i=1}^n X_i Y_i - \hat{\beta}_0 \bar{X} - \hat{\beta}_1 \cdot \frac{1}{n} \sum_{i=1}^n X_i^2 = 0, \quad (5)$$

using $\frac{1}{n} \sum_{i=1}^n Y_i = \bar{Y}$, $\frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$ in (4), and $\frac{1}{n} \sum_{i=1}^n \hat{\beta}_0 X_i = \hat{\beta}_0 \bar{X}$, $\frac{1}{n} \sum_{i=1}^n \hat{\beta}_1 X_i^2 = \hat{\beta}_1 \cdot \frac{1}{n} \sum_{i=1}^n X_i^2$ in (5) (rules (R1),(R3)).

Step 4 — solve (4) for $\hat{\beta}_0$. Equation (4) rearranges immediately to

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Step 5 — substitute into (5) and solve for $\hat{\beta}_1$. Plugging $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ into (5),

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n X_i Y_i - (\bar{Y} - \hat{\beta}_1 \bar{X}) \bar{X} - \hat{\beta}_1 \cdot \frac{1}{n} \sum_{i=1}^n X_i^2 = 0 \\ \Rightarrow & \frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y} + \hat{\beta}_1 \bar{X}^2 - \hat{\beta}_1 \cdot \frac{1}{n} \sum_{i=1}^n X_i^2 = 0. \end{aligned}$$

Collecting the $\hat{\beta}_1$ terms on one side,

$$\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y} = \hat{\beta}_1 \left[\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 \right],$$

so, provided $\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 \neq 0$ (i.e. X is not constant across the sample),

$$\hat{\beta}_1 = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2}.$$

Step 6 — rewrite in deviation-from-mean form, using Question 4. Question 4 shows $\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - n \bar{X} \bar{Y}$ and $\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n \bar{X}^2$; dividing both sides of each identity by n gives exactly the numerator and denominator above. Hence

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X},}$$

where $s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$ is the sample covariance of X, Y and $s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is the sample variance of X (the common factor $1/(n-1)$ cancels in the ratio, so the deviation-sum form and the s_{XY}/s_X^2 form are identical).

Second-order condition. The point solving (4)–(5) is a minimum, not a maximum or saddle, because the Hessian of S is

$$H = \begin{pmatrix} \frac{\partial^2 S}{\partial b_0^2} & \frac{\partial^2 S}{\partial b_0 \partial b_1} \\ \frac{\partial^2 S}{\partial b_1 \partial b_0} & \frac{\partial^2 S}{\partial b_1^2} \end{pmatrix} = 2 \begin{pmatrix} n & \sum_{i=1}^n X_i \\ \sum_{i=1}^n X_i & \sum_{i=1}^n X_i^2 \end{pmatrix}.$$

The leading principal minor is $2n > 0$, and

$$\det H = 4 \left[n \sum_{i=1}^n X_i^2 - \left(\sum_{i=1}^n X_i \right)^2 \right] = 4n \sum_{i=1}^n (X_i - \bar{X})^2 \geq 0,$$

using the same expansion as the variance identity in Question 4; the determinant is strictly positive whenever the X_i are not all identical. Both leading principal minors are then positive, so H is positive definite and the unique stationary point $(\hat{\beta}_0, \hat{\beta}_1)$ is a strict global minimum of $S(b_0, b_1)$ (which is a convex, in fact quadratic, function of (b_0, b_1)).

Method 2: Centred reparameterisation (decouples the FOCs)

Write the same population line through its value at $X = \bar{X}$: for a constant α ,

$$Y_i = \alpha + \beta_1(X_i - \bar{X}) + u_i, \quad i = 1, \dots, n,$$

which is algebraically the same model as $Y_i = \beta_0 + \beta_1 X_i + u_i$ with $\alpha = \beta_0 + \beta_1 \bar{X}$. Minimise

$$T(a, b_1) = \sum_{i=1}^n [Y_i - a - b_1(X_i - \bar{X})]^2$$

over (a, b_1) .

FOC in a .

$$\frac{\partial T}{\partial a} = -2 \sum_{i=1}^n [Y_i - \hat{a} - \hat{b}_1(X_i - \bar{X})] = 0 \implies \sum_{i=1}^n Y_i - n\hat{a} - \hat{b}_1 \sum_{i=1}^n (X_i - \bar{X}) = 0.$$

But $\sum_{i=1}^n (X_i - \bar{X}) = \sum_{i=1}^n X_i - n\bar{X} = n\bar{X} - n\bar{X} = 0$ identically, *regardless of $\hat{b}_1$* . So the $\hat{b}_1$ term vanishes on its own and the equation collapses to $\sum_{i=1}^n Y_i - n\hat{a} = 0$, i.e.

$$\boxed{\hat{a} = \bar{Y}}$$

directly, with no need to know $\hat{b}_1$ first: centring the regressor *decouples* the two FOCs, which is the genuine difference from Method 1 (there, $\hat{\beta}_0$ could only be isolated in terms of the still-unknown $\hat{\beta}_1$, forcing the substitution step of Step 5 above).

FOC in b_1 .

$$\frac{\partial T}{\partial b_1} = -2 \sum_{i=1}^n [Y_i - \hat{a} - \hat{b}_1(X_i - \bar{X})](X_i - \bar{X}) = 0.$$

Substituting $\hat{a} = \bar{Y}$,

$$\sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X}) = \hat{b}_1 \sum_{i=1}^n (X_i - \bar{X})^2 \implies \hat{b}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \hat{\beta}_1,$$

the same slope as Method 1.

Recovering $\hat{\beta}_0$. Matching $\alpha + \beta_1(X_i - \bar{X}) = \alpha - \beta_1\bar{X} + \beta_1 X_i$ against $\beta_0 + \beta_1 X_i$ gives $\beta_0 = \alpha - \beta_1\bar{X}$, so

$$\hat{\beta}_0 = \hat{a} - \hat{\beta}_1\bar{X} = \bar{Y} - \hat{\beta}_1\bar{X},$$

identical to Step 4 of Method 1. Both routes agree, confirming

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X}, \quad \hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

Interpretation. $\hat{\beta}_1$ is the sample covariance between X and Y divided by the sample variance of X — it measures how strongly Y co-moves with X in the sample, scaled by how spread out X is. $\hat{\beta}_0$ is then whatever intercept makes the fitted line pass through the point of sample means $(\bar{X}, \bar{Y})$ (this exact fact is used, and proved, in Question 5(c)).

Technique note. The substitute-then-solve route (Method 1) is the one printed in the lecture note and is the default to reproduce on an exam; the centred reparameterisation (Method 2) is worth knowing because it is the same trick used later (Lecture 4's $\bar{Y}$ variance derivation, and multiple regression's Frisch–Waugh partialling-out) to isolate one coefficient without solving a joint system.

Question 4 (Two summation identities)

Problem

Show the following holds:

$$\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2}, \quad \text{where } \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

(Hint: show $\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$ and $\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}$.) (Tutorial Question Set No. 2, Q4.)

Solution

Convention note. The question sheet displays (4) as a single fraction: the printed numerator equation is

$$\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y}, \quad (4\text{-num, as printed})$$

and the printed denominator equation is

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2. \quad (4\text{-den, as printed})$$

Read as two stand-alone identities, both are off by a factor of n : the left side is a raw sum over n terms, while the right side already carries a $1/n$. For example (4-den) as printed would require $\sum_{i=1}^n (X_i - \bar{X})^2$ (which scales with n for a fixed underlying distribution) to equal $\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2$ (which does *not* scale with n in the same way) — these cannot both be literally true for every n . What *is* true, and is what (4-num) and (4-den) are really asserting when read as *one* fraction equal to another fraction (numerator over numerator, denominator over denominator, as the question sheet lays them out), is the *ratio* identity:

$$\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2},$$

i.e. the two equations share the *same* missing factor of n , which cancels once numerator is divided by denominator. The official Tut 2 slide is careful about exactly this: its own Question 4 solution proves *only* the sum-form identities

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2, \quad \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}$$

(no $1/n$ anywhere), labelled (7) and (8) on the slide, and only afterwards remarks “divide both numerator and denominator by n ” to reach the fraction printed as (4) on the question sheet. The slide never asserts the $1/n$ form as a free-standing equality — it only ever appears already paired inside the ratio, where the extra n is harmless. We follow that same convention below: we prove the n -form identities in full, and then divide top and bottom by n only at the very last step, to land exactly on the ratio printed in the question.

Method 1: Direct expansion

Part (a): the X -variance identity. Expand the square using (R1)–(R2) and $\bar{X}$ constant across the sum:

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2) = \sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + \sum_{i=1}^n \bar{X}^2.$$

Since $\sum_{i=1}^n X_i = n\bar{X}$ (definition of $\bar{X}$) and $\sum_{i=1}^n \bar{X}^2 = n\bar{X}^2$ (rule (R3), $\bar{X}^2$ constant),

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - 2\bar{X}(n\bar{X}) + n\bar{X}^2 = \sum_{i=1}^n X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 = \boxed{\sum_{i=1}^n X_i^2 - n\bar{X}^2}.$$

Part (b): the (X, Y) -covariance identity. Expand the product similarly:

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) &= \sum_{i=1}^n (X_i Y_i - X_i \bar{Y} - \bar{X} Y_i + \bar{X} \bar{Y}) \\ &= \sum_{i=1}^n X_i Y_i - \bar{Y} \sum_{i=1}^n X_i - \bar{X} \sum_{i=1}^n Y_i + \sum_{i=1}^n \bar{X} \bar{Y}. \end{aligned}$$

Using $\sum_{i=1}^n X_i = n\bar{X}$, $\sum_{i=1}^n Y_i = n\bar{Y}$, and $\sum_{i=1}^n \bar{X} \bar{Y} = n\bar{X} \bar{Y}$ (rule (R3)),

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) &= \sum_{i=1}^n X_i Y_i - \bar{Y}(n\bar{X}) - \bar{X}(n\bar{Y}) + n\bar{X} \bar{Y} \\ &= \sum_{i=1}^n X_i Y_i - n\bar{X} \bar{Y} - n\bar{X} \bar{Y} + n\bar{X} \bar{Y} = \boxed{\sum_{i=1}^n X_i Y_i - n\bar{X} \bar{Y}}. \end{aligned}$$

Method 2: Using $\sum_{i=1}^n (X_i - \bar{X}) = 0$

A shortcut avoids expanding both cross-terms in (a)/(b). Since $\bar{X}$ is a constant,

$$\sum_{i=1}^n (X_i - \bar{X}) = \sum_{i=1}^n X_i - n\bar{X} = n\bar{X} - n\bar{X} = 0.$$

Part (b) via the shortcut. Because $\bar{Y}$ is a constant, $\sum_{i=1}^n (X_i - \bar{X})\bar{Y} = \bar{Y} \sum_{i=1}^n (X_i - \bar{X}) = \bar{Y} \cdot 0 = 0$. Hence

$$\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n (X_i - \bar{X})Y_i - \sum_{i=1}^n (X_i - \bar{X})\bar{Y} = \sum_{i=1}^n (X_i - \bar{X})Y_i - 0 = \sum_{i=1}^n (X_i - \bar{X})Y_i.$$

Now expand this single remaining cross-term:

$$\sum_{i=1}^n (X_i - \bar{X})Y_i = \sum_{i=1}^n X_i Y_i - \bar{X} \sum_{i=1}^n Y_i = \sum_{i=1}^n X_i Y_i - \bar{X}(n\bar{Y}) = \sum_{i=1}^n X_i Y_i - n\bar{X} \bar{Y},$$

matching Method 1(b).

Part (a) via the shortcut. Take $Y_i = X_i$ in the identity just proved (or repeat the same argument with $\bar{X}$ in place of $\bar{Y}$): $\sum_{i=1}^n (X_i - \bar{X})\bar{X} = \bar{X} \sum_{i=1}^n (X_i - \bar{X}) = 0$, so

$$\begin{aligned}\sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i - \bar{X})X_i - \sum_{i=1}^n (X_i - \bar{X})\bar{X} = \sum_{i=1}^n (X_i - \bar{X})X_i - 0 \\ &= \sum_{i=1}^n X_i^2 - \bar{X} \sum_{i=1}^n X_i = \sum_{i=1}^n X_i^2 - n\bar{X}^2,\end{aligned}$$

matching Method 1(a).

Combining, and recovering the printed fraction. Dividing both established identities by n ,

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y}, \quad \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2.$$

Since a common factor of $1/n$ appears in *both* the numerator and the denominator of the left-hand ratio, it cancels when the ratio is formed, so

$$\boxed{\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2}}.$$

This is exactly the identity used in Question 3, Step 6 to convert the “ $1/n$ -sum” form of $\hat{\beta}_1$ into the deviation-from-mean form $\hat{\beta}_1 = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) / \sum_{i=1}^n (X_i - \bar{X})^2$.

Interpretation. Both identities say the same structural thing: a sum of squared (or cross-multiplied) *deviations from the mean* can always be rewritten as a sum of squared (or cross-multiplied) *raw* values, minus a correction term built from the means. This “raw minus mean-correction” form is exactly the computational shortcut (sometimes called the “machine formula” for variance/covariance) used when calculating s_X^2 or s_{XY} by hand or in a single pass through the data, without first computing every deviation $X_i - \bar{X}$.

Technique note. Any identity of the form $\sum_{i=1}^n (X_i - \bar{X}) \cdot (\text{anything constant across } i) = 0$ is the single most reusable trick in this course’s algebra: it is exactly how $\sum_{i=1}^n \hat{u}_i X_i = 0$ combines with $\sum_{i=1}^n \hat{u}_i = 0$ later, and it is what collapses Method 2’s FOC in a (Question 3) to a single line.

Question 5 (Class size and test scores: numerical predictions)

Problem

Suppose that a researcher, using data on class size (CS) and average test scores from 50 third-grade classes, estimates the OLS regression

$$\widehat{\text{TestScore}} = 640.3 - 4.93 \times \text{CS}.$$

- (a) A classroom has 25 students. What is the OLS regression's prediction for that classroom's average test score?
- (b) Last year a classroom had 21 students, and this year it has 24 students. What is the OLS regression's prediction for the change in the classroom average test score?
- (c) The sample average class size across the 50 classrooms is 22.8. What is the sample average of the test scores across the 50 classrooms? (*Hint: review the formulas of the OLS estimators.*)

(Tutorial Question Set No. 2, Q5.)

Solution

Throughout, $\hat{\beta}_0 = 640.3$ and $\hat{\beta}_1 = -4.93$ are the OLS estimates (already-computed numbers, i.e. estimates rather than the random variables $\hat{\beta}_0, \hat{\beta}_1$ of Question 2), CS is class size (students per classroom, a count), and $\widehat{\text{TestScore}}$ is the fitted average test score (points, on whatever scale the underlying test used).

Part (a): prediction at CS = 25

Method: plug into the fitted line.

$$\widehat{\text{TestScore}} = 640.3 - 4.93 \times 25 = 640.3 - 123.25 = 517.05.$$

(Carrying the multiplication through exactly: $4.93 \times 25 = 123.25$.)

$$\widehat{\text{TestScore}} = 640.3 - 123.25 = 517.05.$$

Interpretation. A classroom with 25 students is predicted to average a test score of 517.05 points. This is a prediction *along the fitted line*, not a claim about any specific classroom in the data (which will scatter around the line by its residual $\hat{u}_i$).

Part (b): predicted change from CS = 21 to CS = 24

Method 1: predict both years, then subtract

$$\widehat{\text{TestScore}}_{\text{last year}} = 640.3 - 4.93 \times 21 = 640.3 - 103.53 = 536.77,$$

$$\widehat{\text{TestScore}}_{\text{this year}} = 640.3 - 4.93 \times 24 = 640.3 - 118.32 = 521.98.$$

The predicted change is

$$\widehat{\Delta \text{TestScore}} = 521.98 - 536.77 = -14.79.$$

$$\boxed{\widehat{\Delta \text{TestScore}} = -14.79 \text{ points.}}$$

Method 2: use $\hat{\beta}_1$ as a marginal effect directly

Because the fitted line is linear, its slope $\hat{\beta}_1$ is the predicted change in $\widehat{\text{TestScore}}$ per unit change in CS, for *any* starting point: for two class sizes CS_1, CS_2 ,

$$\widehat{\text{TestScore}}_2 - \widehat{\text{TestScore}}_1 = [\hat{\beta}_0 + \hat{\beta}_1 \text{CS}_2] - [\hat{\beta}_0 + \hat{\beta}_1 \text{CS}_1] = \hat{\beta}_1 (\text{CS}_2 - \text{CS}_1) = \hat{\beta}_1 \Delta \text{CS}.$$

Here $\Delta \text{CS} = 24 - 21 = 3$ additional students, so

$$\widehat{\Delta \text{TestScore}} = \hat{\beta}_1 \times \Delta \text{CS} = (-4.93) \times 3 = -14.79,$$

matching Method 1 without computing either predicted level.

$$\boxed{\widehat{\Delta \text{TestScore}} = \hat{\beta}_1 \Delta \text{CS} = -4.93 \times 3 = -14.79 \text{ points.}}$$

Interpretation. Adding three students to the classroom (a class-size *increase*, $\Delta \text{CS} = +3$) is predicted to *lower* the average test score by 14.79 points (-4.93 points per additional student, times 3 students). The sign is consistent with $\hat{\beta}_1 < 0$: larger classes are associated with lower average test scores in this sample.

Part (c): sample average test score, given mean CS = 22.8

Step 1 — why the OLS line passes through $(\overline{\text{CS}}, \overline{\text{TestScore}})$. By Question 3, the OLS intercept satisfies

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \quad \Longleftrightarrow \quad \bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X},$$

i.e. (with $X = \text{CS}$, $Y = \text{TestScore}$) the sample mean of Y equals the fitted value of the OLS line evaluated exactly at the sample mean of X . Equivalently: substituting $X_i = \bar{X}$ into the fitted line $\hat{\beta}_0 + \hat{\beta}_1 X_i$ gives $\hat{\beta}_0 + \hat{\beta}_1 \bar{X} = \bar{Y}$ by the rearrangement above — the fitted regression line always passes through the point of means $(\bar{X}, \bar{Y})$, for *any* data set, because that is precisely how $\hat{\beta}_0$ was solved for in Question 3, Step 4.

Step 2 — apply this with the given numbers. Here $\hat{\beta}_0 = 640.3$, $\hat{\beta}_1 = -4.93$, and $\overline{\text{CS}} = 22.8$, so

$$\overline{\text{TestScore}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \overline{\text{CS}} = 640.3 - 4.93 \times 22.8.$$

Computing $4.93 \times 22.8 = 112.404$,

$$\overline{\text{TestScore}} = 640.3 - 112.404 = 527.896.$$

$$\boxed{\overline{\text{TestScore}} = 527.896 \text{ points.}}$$

Interpretation. No new information about individual classrooms is needed to answer part (c): because the fitted line necessarily runs through $(\bar{CS}, \bar{\text{TestScore}})$, plugging the sample mean class size into the fitted line *is* the sample mean test score. This is the practical payoff of Question 3's identity $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$: it lets you recover $\bar{Y}$ from $\hat{\beta}_0, \hat{\beta}_1, \bar{X}$ alone, without ever seeing the raw TestScore_i data.

Beyond the slides. The official Tut 2 slide states the numerical answer to (c) but only cites “review the formulas of the OLS estimators” as justification. Step 1 above supplies the missing one-line proof (rearranging $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$) that the answer follows *exactly*, not just approximately, from the definition of $\hat{\beta}_0$ — this is worth stating explicitly, since it is a fact ($\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$) that is reused whenever R^2 , ESS, or TSS are computed in later lectures.

Numerical check (verified in Python). All four arithmetic results above were re-computed independently:

```
>>> 640.3 - 4.93*25
517.05
>>> 640.3 - 4.93*21, 640.3 - 4.93*24, (640.3 - 4.93*24) - (640.3 - 4.93*21)
(536.77, 521.98, -14.790000000000006)
>>> -4.93*3
-14.79
>>> 640.3 - 4.93*22.8
527.8960000000001
```

(the $\sim 10^{-13}$ discrepancies are ordinary floating-point rounding and confirm 517.05, 536.77, 521.98, -14.79 , 527.896 exactly).

Method Map

Q.	Core topic	Methods included
1	Idea of OLS / minimisation problem	Population model $\rightarrow$ candidate line $\rightarrow$ prediction error $\rightarrow$ sum-of-squares objective $\rightarrow$ arg min; redrawn scatter/line/error figure.
2	Notation (β_0, β_1) vs. (b_0, b_1) vs. $(\hat{\beta}_0, \hat{\beta}_1)$	Trace each symbol to where it first appears (model, objective, minimiser); side-by-side comparison table.
3	Deriving $\hat{\beta}_0, \hat{\beta}_1$ from the FOCs	Chain-rule partials $\rightarrow$ normal equations $\rightarrow$ solve $\hat{\beta}_0$ then substitute for $\hat{\beta}_1$; second-order condition via the Hessian; centred reparameterisation that decouples the FOCs.
4	Two summation identities	Direct expansion of the square/cross-product; shortcut via $\sum_{i=1}^n (X_i - \bar{X}) = 0$; convention note on the sheet's missing $1/n$ factor.
5	Class-size/test-score predictions	Direct substitution into the fitted line; $\hat{\beta}_1 \times \Delta X$ as a marginal-effect shortcut; the fitted line through $(\bar{X}, \bar{Y})$.

Technique Summary

- Three layers of notation.** β (unhatted) is the unknown population truth; a plain Roman letter is a dummy argument of an optimisation problem; $\hat{\beta}$ (hatted) is the solution of that optimisation, hence a random variable whose numerical value on one sample is an “estimate.”
- FOCs are two linear equations in two unknowns.** Chain-rule differentiation of a sum of squares always produces $-2 \times (\text{sum of residual-like terms}) = 0$; dividing by $-2n$ turns them into sample-mean statements, which is why $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ appears so directly.
- Centring decouples FOCs.** Re-parameterising the model as $Y_i = \alpha + \beta_1(X_i - \bar{X}) + u_i$ uses $\sum_{i=1}^n (X_i - \bar{X}) = 0$ to solve $\hat{\alpha} = \bar{Y}$ immediately, without waiting on $\hat{\beta}_1$; the same $\sum_{i=1}^n (X_i - \bar{X}) = 0$ trick collapses cross-terms throughout Question 4.
- Raw-sum vs. deviation-sum identities.** Any $\sum_{i=1}^n (X_i - \bar{X})Y_i$ -type sum equals $\sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}$; check any printed version of this identity for a consistent power of n on both sides before using it as a “fact.”
- The fitted line always passes through $(\bar{X}, \bar{Y})$.** A direct consequence of how $\hat{\beta}_0$ is solved for; it lets you recover $\bar{Y}$ (or predict at the mean) from $\hat{\beta}_0, \hat{\beta}_1, \bar{X}$ alone.

Final Answers

- OLS estimators solve $(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{b_0, b_1} \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2$ (see Figure 1 for the geometric picture of the squared prediction errors being minimised).

2. β_0, β_1 : true unknown population parameters. b_0, b_1 : generic candidate values (arguments of the minimisation). $\hat{\beta}_0, \hat{\beta}_1$: the minimisers – OLS estimators, random variables that are functions of the sample and vary from sample to sample; their numerical values on one sample are the estimates.
3. $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$, $\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$; the stationary point is a strict minimum because the Hessian of the sum-of-squares objective is positive definite.
4. $\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$ and $\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}$ (sum forms, no $1/n$); dividing both by n and forming the ratio recovers the fraction printed as (4) on the question sheet.
5. (a) 517.05; (b) predicted change = -14.79 points (equivalently $\hat{\beta}_1 \times 3 = -14.79$); (c) sample mean test score = 527.896 points, because $\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$.